

University AI Operating Model

An intelligence layer over Edugate (SIS) and Edulearn (LMS)

Outcome briefing · 9 AI agents · End-to-end student journey

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What we will cover today

01

Strategic Context

Why an AI layer, and why now

02

Operating Model

Student journey end-to-end with embedded AI

03

Architecture

Five layers from user to data

04

The 9 AI Agents

Specifications and stakeholder value

The problem we are solving

The university operates reactively. Risk surfaces too late, faculty are overloaded, and students churn through avoidable friction.

Pain Point

students change major within first year

MIS-ENROLLMENT

Pain Point

registrations fail prerequisite/credit checks

OPERATIONAL DRAG

Pain Point

avg. lead time before at-risk students are flagged

LATE DETECTION

Pain Point

on-time graduation rate

OUTCOME GAP

THE OPPORTUNITY

Edugate and Edulearn already hold the data. We do not need to replace them — we need an intelligence layer that turns that data into timely decisions for students, faculty, and administrators.

Five design principles

1	Layer, do not replace	Edugate (SIS) and Edulearn (LMS) remain the systems of record. The AI layer reads, recommends, and acts through them.
2	AI recommends, humans decide	Every consequential decision — admission, grading, exception — remains owned by a named human.
3	Multi-stakeholder by design	Each agent must serve students, faculty, and administration simultaneously. No agent optimizes for one at the expense of another.
4	Embedded at decision points	Agents trigger at specific moments in the journey, not as generic chatbots or dashboards.
5	Measured on outcomes and trust	Success means both better student outcomes and high human acceptance of recommendations.

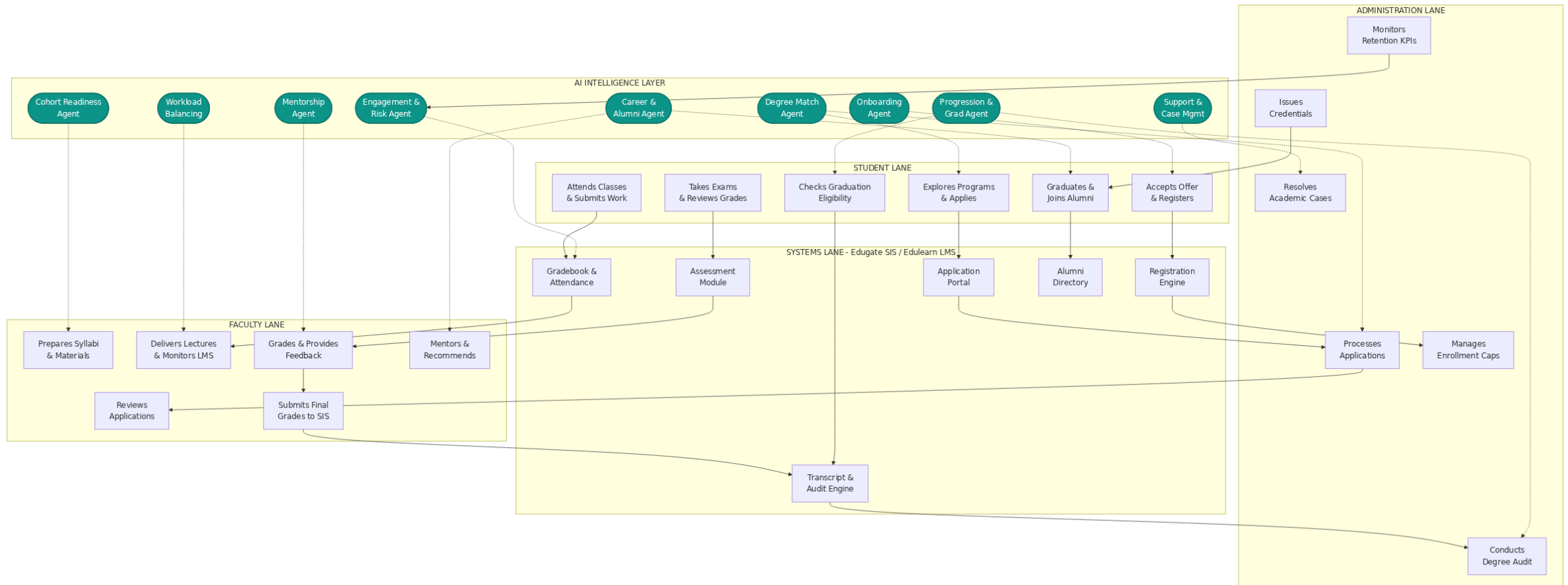
The student journey, end-to-end



AI AGENTS TRIGGERED AT EACH STAGE

Awareness → Admission	— Degree Match & Pathway Intelligence
Enrollment	— Transition, Onboarding & Registration
Pre-semester (Faculty)	— Cohort Readiness & Teaching Preparation
Academic Journey (continuous)	— Engagement & Early Risk Detection · Academic Mentorship · Workload Balancing
Continuous (all stages)	— Student Support & Case Management
Every Semester · Graduation	— Progression & Graduation Intelligence
Year 2 → Alumni	— Career & Alumni Intelligence

Swimlane: who does what



Five lanes — Student, Faculty, Administration, Systems (SIS/LMS), and the AI Layer — show how responsibilities interact at each stage.

Five-layer architecture



Telemetry feedback ↑

Flow ↓

At a glance

1

Degree Match & Pathway

Right student → right program
AWARENESS → ADMISSION

2

Onboarding & Registration

Frictionless first semester
ENROLLMENT

3

Cohort Readiness & Teaching Prep

Faculty know their cohort
PRE-SEMESTER

4

Engagement & Risk Detection

Catch risk by week 2
CONTINUOUS

5

Workload Balancing

No faculty burnout
PRE-SEM + MID-TERM

6

Academic Mentorship

Right mentor, right time
TRIGGERED BY AGENT 4

7

Student Support & Case Mgmt

One door, right routing
CONTINUOUS

8

Progression & Graduation

Off-track? Know in semester 1
EVERY SEMESTER

9

Career & Alumni

Graduate to outcome
YEAR 2 → ALUMNI

Degree Match & Pathway Intelligence

AWARENESS → APPLICATION → ADMISSION

Recommends optimal degree pathways at admission and validates course selections against curriculum rules.

INPUTS · from SIS / LMS

- Applicant profile & academic history (SIS)
- Program prerequisites & capacity
- Historical fit & success outcomes

ACTIONS & OUTPUTS

- ▶ Recommends best-fit programs with rationale
- ▶ Flags weak fit before admission
- ▶ Auto-validates first-semester course list

STUDENT

Applies to a program where they will succeed — not just one they qualify for.

FACULTY

Receives a cohort that is academically prepared for the curriculum.

ADMINISTRATION

Reduces mis-enrollment rework and downstream major changes.

SIGNATURE KPIs

Pathway recommendation accuracy

Mis-enrollment in year 1

Transition, Onboarding & Registration

ENROLLMENT

Automates onboarding tasks and validates first-semester registration against prerequisites and credit limits.

INPUTS · from SIS / LMS

- New-student registration data (SIS)
- Onboarding checklist status
- Course catalog & prerequisites

ACTIONS & OUTPUTS

- ▶ Auto-validates schedules in real time
- ▶ Surfaces missing onboarding tasks
- ▶ Routes exceptions to registrar with context

STUDENT

Registers in hours, not days — with no preventable errors.

FACULTY

Class lists are accurate from day 1.

ADMINISTRATION

Registrar handles only true exceptions, not routine validation.

SIGNATURE KPIs

Registration error rate

Registration cycle time

Cohort Readiness & Teaching Preparation

PRE-SEMESTER (FACULTY)

Profiles incoming cohorts and surfaces preparation gaps to faculty before semester start.

INPUTS · from SIS / LMS

- Cohort academic background (SIS)
- Prior LMS engagement patterns
- Prerequisite gap analysis

ACTIONS & OUTPUTS

- ▶ Delivers cohort profile to faculty 7+ days pre-term
- ▶ Recommends syllabus adjustments
- ▶ Pre-enrolls students into remediation

STUDENT

Enters the term ready, with gaps addressed before week 1.

FACULTY

Walks in knowing the cohort — not discovering it in week 3.

ADMINISTRATION

Reduces week-1 drops and mid-term remediation cost.

SIGNATURE KPIs

Prerequisite gap remediation

Week-1 attrition

Engagement & Early Risk Detection

ACADEMIC JOURNEY (CONTINUOUS)

Monitors LMS telemetry and SIS grades to detect disengagement and at-risk patterns by week 2.

INPUTS · from SIS / LMS

- LMS login, content & assignment telemetry
- Grade trajectory (SIS)
- Attendance & financial holds

ACTIONS & OUTPUTS

- ▶ Issues tiered alerts (Yellow/Orange/Red)
- ▶ Differentiates "disengaged" vs "struggling"
- ▶ Routes to Mentorship or Support agent

STUDENT

Gets help when intervention still works — not after failing.

FACULTY

Sees who needs intervention and why, not a wall of names.

ADMINISTRATION

Retention rises; failure-driven appeals decline.

SIGNATURE KPIS

Risk detection lead time

Pass-rate uplift (intervened vs. control)

Workload Balancing

PRE-SEMESTER + MID-TERM

Analyzes faculty teaching, advising, research, and grading load to recommend rebalancing.

INPUTS · from SIS / LMS

- Faculty assignments (HR + SIS)
- Grading & advising activity (LMS)
- Contractual load definitions

ACTIONS & OUTPUTS

- ▶ Flags overload before term starts
- ▶ Proposes section reassignments to chair
- ▶ Tracks grading turnaround in real time

STUDENT

Faster grading turnaround and more available faculty.

FACULTY

Workload is fair and visible; burnout is preventable.

ADMINISTRATION

Chairs make data-backed staffing decisions.

SIGNATURE KPIS

Overload incidents per term

Grading turnaround time

Academic Mentorship & Intervention

ACADEMIC JOURNEY (TRIGGERED BY AGENT 4)

Matches at-risk or high-potential students with mentors and structured intervention plans.

INPUTS · from SIS / LMS

- Risk-detection signals (Agent 4)
- Mentor availability & expertise
- Historical intervention outcomes

ACTIONS & OUTPUTS

- ▶ Auto-matches student to mentor in days
- ▶ Generates structured intervention plan
- ▶ Tracks plan completion and GPA recovery

STUDENT

Matched to the right mentor — quickly.

FACULTY

Mentoring is targeted, not load-spread.

ADMINISTRATION

Retention and GPA-recovery KPIs improve measurably.

SIGNATURE KPIs

Time to first mentor meeting

GPA recovery (≥0.3 next term)

Student Support & Case Management

CONTINUOUS (ALL STAGES)

Pre-diagnoses non-academic cases (financial, wellness, accessibility) and routes to the correct office with full context.

INPUTS · from SIS / LMS

- Helpdesk tickets & history
- Financial holds (SIS)
- Engagement signals (LMS)

ACTIONS & OUTPUTS

- ▶ Routes to correct office on first try
- ▶ Pre-attaches context and history
- ▶ Detects repeat-issue patterns

STUDENT

One door, fast resolution — no re-explaining the same story.

FACULTY

Faculty step out of the role of triaging non-academic issues.

ADMINISTRATION

Caseload halves; first-contact resolution rises sharply.

SIGNATURE KPIS

First-contact resolution

Case resolution time

Progression & Graduation Intelligence

EVERY SEMESTER (PROGRESSION)

Continuous degree audit from semester 1; flags off-track students and projects graduation date.

INPUTS · from SIS / LMS

- Completed credits & grades (SIS)
- Curriculum requirements & substitutions
- Cohort progression patterns

ACTIONS & OUTPUTS

- ▶ Continuous degree audit every term
- ▶ Projects graduation date with confidence interval
- ▶ Flags off-track 4+ semesters early

STUDENT

Knows graduation status every term — not in the final year.

FACULTY

Advisors focus on edge cases, not routine audits.

ADMINISTRATION

On-time graduation rises; late-stage exceptions fall.

SIGNATURE KPIS

Off-track detection lead time

On-time graduation rate

Career & Alumni Intelligence

YEAR 2 → GRADUATION → ALUMNI

Aligns coursework, internships, and post-graduation outcomes with the alumni network and labor-market signals.

INPUTS · from SIS / LMS

- Student major, electives, GPA (SIS)
- Labor-market & employer-skill signals
- Alumni outcomes & network (CRM)

ACTIONS & OUTPUTS

- ▶ Recommends career-aligned electives
- ▶ Matches to internships and alumni mentors
- ▶ Surfaces curriculum gaps to chairs

STUDENT

Graduates with a path, a network, and a job offer.

FACULTY

Curriculum stays aligned with the labor market.

ADMINISTRATION

Employment outcomes and alumni engagement rise.

SIGNATURE KPIs

6-month employment rate

Internship placement rate